THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The 10th term of the AP $\frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 4

2. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. Then $AE : AC$ is 2: 1

(a) 7 **(b)** 5 **(c)** 4 **(d)** 6

3. The point $(0, 0)$ is equidistant from $(2, 3)$ and $(6, 1)$ for $x =$ 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 4

4. The 15th term of $8, 13, 18, 23, \ldots$ is: 1

(a) 79 **(b)** 78 **(c)** 77 **(d)** 80

5. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $DB = 6$ cm, $AE = 6$ cm. Then EC (in cm) $=$ 1

(a) 14 **(b)** 13 **(c)** 11 **(d)** 12

6. The distance between $(-1, 2)$ and $(4, 14)$ is: 1

(a) 12 **(b)** 13 **(c)** 14 **(d)** 15

7. The 8th term from the end of the AP with last term 80 and $d = 3$ is: 1

(a) 58 **(b)** 60 **(c)** 59 **(d)** 61

8. $\triangle ABC \sim \triangle DEF$ with areas 18 cm² and 32 cm². If $AB = 9$ cm, then DE (cm) $=$ 1

(a) 13 **(b)** 12 **(c)** 14 **(d)** 11

9. The point which divides the join of $(3, 3)$ and $(9, 12)$ in the ratio 1:2 internally is: 1

(a) $(6, \frac{15}{2})$ **(b)** $(9, 12)$ **(c)** $(7, 9)$ **(d)** $(5, 6)$

10. If $a_n = 3n + 5$, then the common difference is: 1

(a) 4 **(b)** 2 **(c)** 5 **(d)** 3

11. The length of the diagonal of a square of side 5 cm, squared, is: 1

(a) 49 **(b)** 50 **(c)** 52 **(d)** 51

12. The area of the triangle with vertices $(0, 0)$, $(8, 0)$ and $(0, 5)$ is: 1

(a) 21 **(b)** 20 **(c)** 19 **(d)** 22

13. If $S_n = n^2 + 3n$, then $a_1 =$ 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 3

14. A man goes 24 m due east and then 32 m due north. His distance from the start (m) is: 1

(a) 42 **(b)** 40 **(c)** 41 **(d)** 39

15. If the midpoint of $(4, 1)$ and (x, y) is $(7, 2)$, then $x =$ 1

(a) 10 **(b)** 12 **(c)** 11 **(d)** 9

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the coordinates of the point which divides the join of $(4, -3)$ and $(8, 5)$ internally in the ratio $3 : 1$. 2

17. Find the sum of the AP -5, -2, \dots, 28. 2

18. The perimeters of two similar triangles are 20 cm and 40 cm. Find the ratio of their corresponding sides and their areas. 2

19. Find the values of y for which the distance between $P(10, y)$ and $Q(2, -3)$ is 10 units. 2

20. Which term of the AP $8, 11, 14, 17, \ldots$ is 71? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 1 : 2$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

22. In what ratio does the x-axis divide the segment joining $(-4, -6)$ and $(3, 8)$? Find the point of division. 3

23. In an AP, the 3rd term is 11 and the 7th term is 23. Find the AP and its 12th term. 3

24. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 5$ cm, $DB = 3$ cm and $AE = 10$ cm, find EC. 3

OR

Find the value of k for which the points $(2, 1)$, $(4, 5)$ and $(6, k)$ are collinear. 3

25. How many terms are there in the AP 6, 10, 14, \dots, 86? 3

OR

The areas of two similar triangles are 18 cm² and 162 cm². If a side of the first is 3 cm, find the corresponding side of the second. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two similar triangles have perimeters 18 cm and 36 cm. If a side of the smaller is 5 cm, find the corresponding side of the larger and the ratio of their areas. 5

OR

Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 2)$, $B(4, 3)$ and $C(5, 7)$. Find the fourth vertex D. 5

27. How many terms of the AP $9, 17, 25, 33, \ldots$ must be taken so that the sum is 450? 5

OR

State and prove the converse of the Basic Proportionality Theorem. 5

END OF QUESTION PAPER — **Total: 50 Marks**